

Amrita Vishwa Vidyapeetham
Amrita School of Computing
Course Plan and Evaluation Plan

Course code	23ECE285
Course name	DIGITAL ELECTRONICS LABORATORY
Academic year	2025 – 26
Program	B.Tech CSE
Semester	Third (Odd)
Year of admission & Batch	2024
No. of students	A(63), B(60), C(62)
Faculty Advisors	
Faculty	Ms Sangita Khare (A & B), Dr Kavya Sai(C section)
L T P CR	3 0 0 3
Total number of hours planned	14 hours

Course Objectives

- To provide hands-on experience in realizing simple logic expressions
- To demonstrate the power of logic function optimization
- To enable the implementation of combinational and sequential circuits.

Course Outcomes

- CO1: Use datasheets & simulation tools effectively
- CO2: Realize simple logic circuits
- CO3: Design & implement combinational circuits
- CO4: Design & implement sequential circuits

CO-PO Mapping

PO/ PSO CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3				3				2	2		2		2
CO2	3								2	2				2
CO3	3	3	2						2	2		2		2
CO4	3	3	2						2	2		2		2

Syllabus

1. Logic Gates: Implement logic gates using NAND / NOR.
2. Boolean functions: using logic gates
3. Combinational circuits: Mux, De-Mux, Code Converters, Adders & Subtractors
4. Sequential Circuits: Flip-Flops, Counters, State Machines

Experiments

1. Verification of Basic Logic Gates.
2. Realization of Basic Gates using Universal Logic Gates.
3. Simplification and Realization of a given Boolean Expression i) Using basic gates ii) SOP Using NAND gates only iii) SOP Using NOR gates only iv) POS Using NAND gates only v) POS Using NOR gates only and vi) Compare and analyze the above implementations
4. Design and verification of Adders and Subtractors.
5. Design and verification of Parallel Adder / Subtractor.
6. Design and verification of Binary to Gray code converter and vice versa.
7. Design and verification of BCD to Excess-3 code converter and vice versa.
8. Design and verification of 2-bit Magnitude Comparator.
9. Design and verification of Multiplexers
10. Implementation and verification of Half adder, full adder, half subtractor and full subtractor using multiplexers.
11. Design and verification of Flip-flops (D, T and JK flipflop).
12. Design and verification of shift Registers.
13. Design and verification of Ring and Johnson Counters.
14. Design and verification of 4-bit asynchronous Up and Down Counters

Text Book(s)

Stephen Brown, Zvonko Vranesic, "Fundamentals of Digital logic with Verilog Design", Tata McGraw Hill Publishing Company Limited, Special Indian Edition, 2007.

Reference(s)

John F. Wakerly, "Digital Design Principles and Practices", Pearson Education, Fourth Edition, 2008.

K A Navas, "Electronic Lab Manual" – Volume 1, Fifth Edition, Prentice Hall of India, 2015.

M Morris Mano and Michael D Ciletti, "Digital Design with Introduction to the Verilog HDL", Pearson Education, Fifth Edition, Fifth Edition, 2015

Course Plan - Detailed Lecture Plan:

Week. No	Topic	Evaluation/ Activities
1	Introduction	
2	Verification of Basic Logic Gates.	
3	Realization of Basic Gates using Universal Logic Gates.	
3	Simplification and Realization of a given Boolean Expression i) Using basic gates ii) SOP Using NAND gates only iii) SOP Using NOR gates only iv) POS Using NAND gates only v) POS Using NOR gates only and vi) Compare and analyze the above implementations	Viva/Assignments 1
4	Design and verification of Adders and Subtractors.	
	Design and verification of Parallel Adder / Subtractor.	
5	Design and verification of Binary to Gray code converter and vice versa.	Viva/Assignments 2
6	Design and verification of BCD to Excess-3 code converter and vice versa.	
7	Design and verification of 2-bit Magnitude Comparator.	
8	-	Mid-Sem Exam
9	Design and verification of Multiplexers	Viva/Assignments 3
	Implementation and verification of Half adder, full adder, half subtractor and full subtractor using multiplexers.	
10	Design and verification of Flip-flops (D, T and JK flipflop).	
11	Design and verification of shift Registers.	Viva/Assignments 4
12	Design and verification of Ring and Johnson Counters.	Missed Mid-Sem Exam
13	Design and verification of 4-bit asynchronous Up and Down Counters	Course End Survey, Faculty Feedback
14		End-Sem Exam

Evaluation Pattern: 80:20

Assessment	Internal	External
Mid Sem Examination	20	-
*Continuous Assessment Laboratory (CAL)	60	-
End Semester	-	20

**CAL includes Lab Evaluations, Assignments*

Evaluation Plan

Component	Number of Evaluations	Weightage
CAL Viva/Assignments	04	20%
(Internal) Lab Assessments	02	40%
Mid-Term Exam (Internal)	01	20%
End Term Exam (External)	01	20%

COs to Evaluations Mapping

Evaluation	Max marks	Weightage	CO Mapping	Tentative schedule
Lab viva / Assignment 1	10	05	CO1, CO2	July last week
Lab viva / Assignment 2	10	05	CO3	Third week of August
Lab viva / Assignment 3	10	05	CO3	Fourth week of september
Lab viva / Assignment 4	10	05	CO4	Third week of october
Lab Evaluation 1	20	20	CO1	Fourth week of August
Lab Evaluation 2	20	20	CO3	Last week of october
Mid term	20	20	CO2	
End term	20	20	CO4	

CO	Mid-Term (20%)			End-Term (20%)	Mark distribution (in %)				Marks
	V1 (05)	V2 (05)	V3 (05)	V4 (05)	LabQuiz1 (20)	LabQuiz2 (20)	Midterm 20	Endterm 20	
CO1	05 (2.5)	-	-	-	20(20)	-	-	-	22.5
CO2	05 (2.5)		-	-	-	-	20 (20)	-	22.5
CO3	-	10 (05)	10 (05)	-	-	20(20)	-	-	30
CO4	-	-	-	10 (05)	-	-	-	20 (20)	25

(Ms Sangita Khare & Dr.Kavya Sai)

Name and Signature of Faculty Name and Signature of Chairperson